

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 11

7 July 2025

Overview:

In this session, we explored some key concepts that form the backbone of understanding modern AI systems. We began by understanding how model performance is evaluated using the confusion matrix. Then, we got introduced to LLMs (Large Language Models), like ChatGPT, and understood how these advanced models are shaping the AI world today. We also discussed the difference between Artificial Intelligence and Machine Learning, clarifying how they are related yet different fields.

Learning Objectives:

- Understand how a confusion matrix helps evaluate classification models.
- Learn what LLMs (Large Language Models) are and how they work.
- Differentiate clearly between AI and ML, and their areas of application

Confusion Matrix:

A confusion matrix is a table used to evaluate the performance of a classification algorithm. It shows the True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). From this matrix, we can check other metrics like accuracy, precision, recall, and F1-score, which help to measure how well a model is performing on both positive and negative classes.

- **True Positive (TP):** The model correctly predicted a positive outcome i.e. the actual outcome was positive.
- **True Negative (TN):** The model correctly predicted a negative outcome i.e. the actual outcome was negative.
- **False Positive (FP):** The model incorrectly predicted a positive outcome i.e. the actual outcome was negative. It is also known as a Type I error.

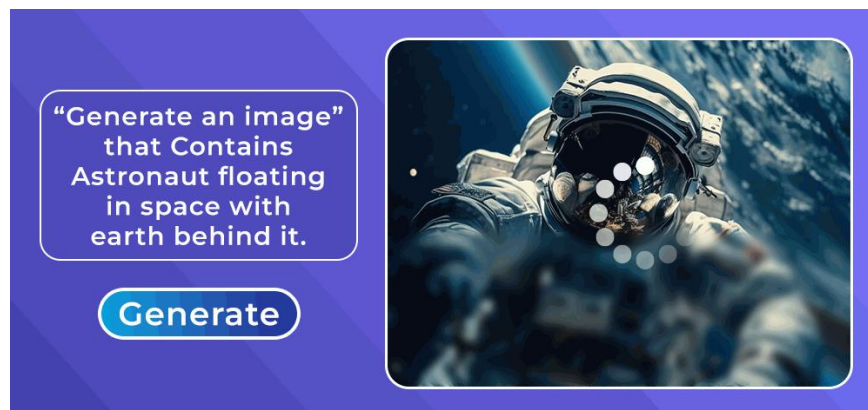
- **False Negative (FN):** The model incorrectly predicted a negative outcome i.e. the actual outcome was positive. It is also known as a Type II error.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

LLM (Large Language Models):

A large language model is a type of artificial intelligence algorithm that applies neural network techniques with lots of parameters to process and understand human languages or text using self-supervised learning techniques. Tasks like text generation, machine translation, summary writing, image generation from texts, machine coding, chat-bots, or Conversational AI are applications of the Large Language Model.

Examples of such LLM models are Chat GPT by open AI, BERT (Bidirectional Encoder Representations from Transformers) by Google, etc.



GPT (Generative Pre-trained Transformer):

- **GPT-1** which was released in 2018 contains 117 million parameters having 985 million words.
- **GPT-2** which was released in 2019 contains 1.5 billion parameters.
- **GPT-3** which was released in 2020 contains 175 billion parameters. Chat GPT is also based on this model as well.
- **GPT-4** model is released in the early 2023 and it is likely to contain trillions of parameters.
- **GPT-4 Turbo** was introduced in late 2023, optimized for speed and cost-efficiency, but its parameter count remains unspecified.

AI vs ML:

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)
Definition	Simulating human intelligence in machines	Training machines to learn from data
Goal	To make smart machines that can think and act	To allow systems to learn and improve from experience
Approach	Broad – includes reasoning, learning, planning, etc.	Focused on using data and algorithms

Conclusion:

Today's session was a great blend of theory and modern application. Understanding the confusion matrix gave us practical tools to evaluate models, while the discussion on LLMs helped us connect classroom learning to real-world AI systems. Clarifying the distinction between AI and ML also gave us a solid foundation going forward. These concepts are crucial in the evolving world of technology, and it was exciting to see how they're connected.